INES-RUHENGERI · FACULTY OF SCIENCES AND INFORMATION TECHNOLOGY

MyTurn

Design and Implementation of an IoT-Based Queue Management and Service-Desk Status System

Your Turn, Made Simple.

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Internet of Things (IoT)
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A numbered line is still a line.

1

Waiting blind

A visitor cannot tell how long the wait is, so they crowd the counter, leave, or come back at the wrong moment.

2

One absent visitor

The desk calls a number that is not there. Without a rule that recovers, everyone behind it waits.

3

No record

A paper list cannot say how long service takes, how many visitors never answered, or who changed the desk.

> The desk needs a number, a live status, and a fair rule for the person who does not answer.

A QR code, a number, and a desk that recovers by itself.

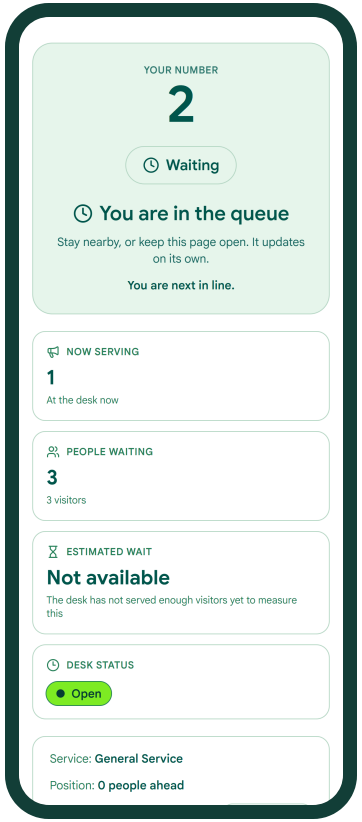
1 **Scan**
The QR code at the door opens the ticket page. No app, no account.

2 **Take a number**
The server issues the next number and a private tracking link.

3 **Follow your turn**
The page shows the called number, the wait and how many are ahead.

4 **Be called**
The display and the phone update together, without a refresh.

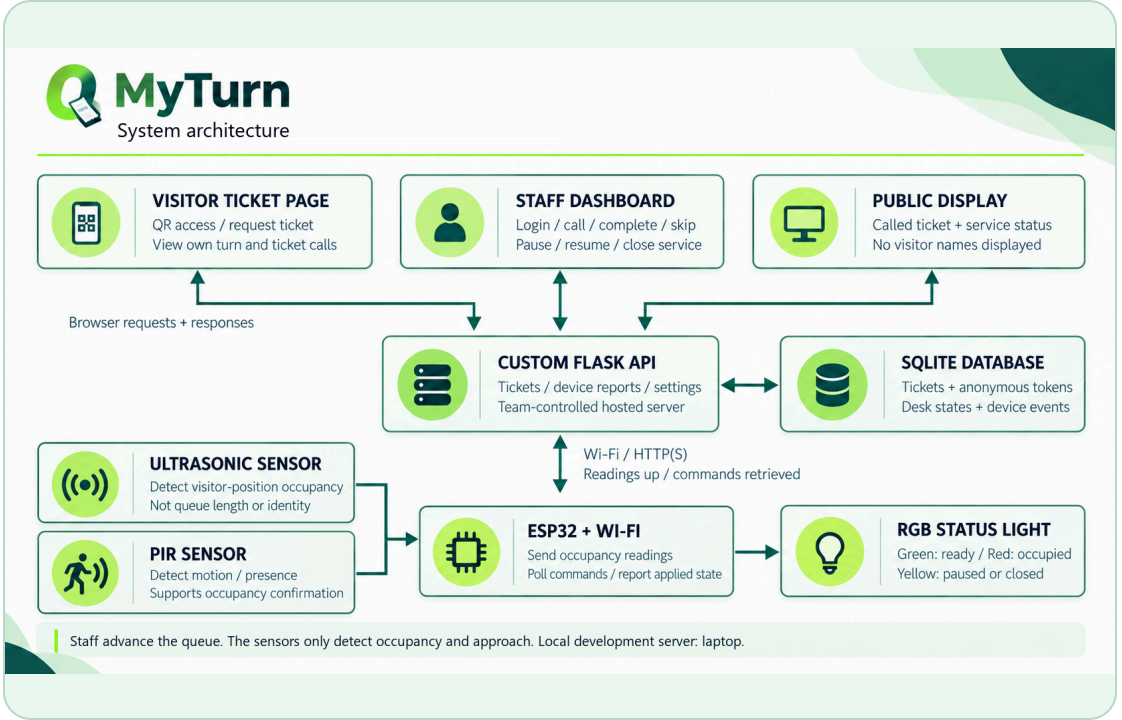
No name · no phone number · no account



The visitor page, built and running

Four parts, one queue.

- **The board measures**
  An ESP32 reads distance and approach, then applies the desk state to an RGB LED.
- **The API decides**
  Flask owns the queue rules, the counter, the accounts and the event log.
- **SQLite remembers**
  Queue, desk, board and accounts in one file, with every change written down.
- **The pages show**
  Next.js: visitor, wall display, staff console and administration.
- **One origin**
  The browser reaches the API through the same origin; the board proves itself with a device key.



ESP32 → Flask API → SQLite → Next.js

What sits at the counter.

ESP32 DevKitC	the controller
HC-SR04 ultrasonic	TRIG 5 · ECHO 18
HC-SR501 motion	GPIO 19
RGB LED	GPIO 25 / 26 / 27

RWF 46,200 estimated prototype budget

The enclosure, the sensor mount and the power supply are part of the same budget.



Wiring at a glance · the full schematic is in the report

The sensors measure a place, not a person.

MEASURED

- ✓ Distance from the HC-SR04, turned into **occupied** or **free**.
- ✓ An approach from the PIR, held for a short window.
- ✓ The LED state the board actually applied, sent back with every report.
- ✓ A median of five distance readings, so one bad echo does not change the state.
- ✓ Each reading is stored with its time, distance, occupancy, approach and applied LED state.

NEVER

- × No counting people in a line.
- × No identity, face, phone or name.
- × No camera, no microphone, no tracking.
- × No guess about who deserves priority.
- × No record of a person, a visit or a conversation.

> A queue system does not need to know who you are. The HC-SR04 alone decides occupancy; the PIR only reports that somebody is walking up.

One line, one call, one fair rule.

1. **The number comes from the server**
   A counter row is updated in the same transaction that inserts the ticket, so two requests cannot share a number.

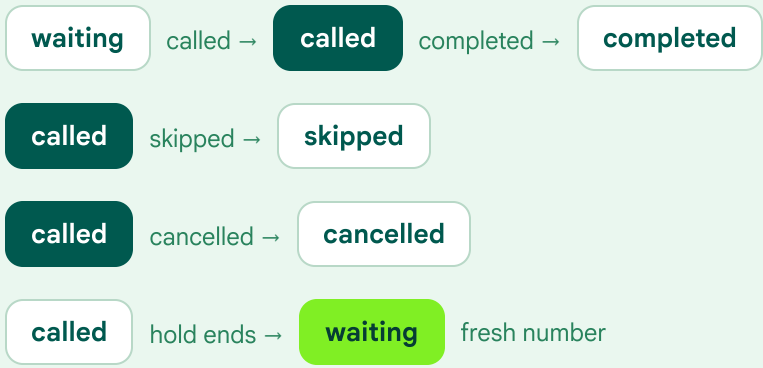
2. **The link is opaque**
   The tracking URL carries a random token, never the ticket number or a sequential id.

3. **First come, first served**
   One call at a time, enforced by a unique index and a transaction.

4. **Nobody blocks the desk**
   After the hold, an unanswered number goes to the back with a fresh number.

5. **Everything is written down**
   Every change is an event with a time and an actor: a person, or the system.

A TICKET'S LIFE



recall restarts the hold; undo takes back the last decision within 10 seconds. The default hold is 60 seconds, and the next slide shows the one case where the board can shorten it.

The board can end an unanswered call early. Only with evidence.

Default hold 60 seconds

0 s 10 s grace 60 s

Without a fresh reading, the full hold always applies.

With evidence 25 seconds



0 s 10 s grace 25 s 60 s

The unused part of the hold is given back to the queue.

grace · counter empty · full hold · not used

Fresh board · **No occupancy** · **No approach** · **Past the grace** · **Otherwise: full hold**

The server decides. The board only reports. An offline board, a missing reading, an occupied counter or an approach keeps the full hold, so absence of data is never treated as absence of a person.

Why a floor? A visitor who is walking in needs time to reach the desk. Ten seconds of grace, then a floor of 25, stops the rule from firing before they arrive.

A wait you can trust, or no number at all.

3

served tickets before the page offers an estimate.
Below that it says “Not available”.

MYTURN_ETA_MIN_SAMPLES, not a guess

estimate = measured average × tickets ahead

1 Measured, not modelled

Each sample is the time from **called_at** to **closed_at** of a ticket that was actually served.

2 The sample count is visible

The visitor sees the average and how many tickets it came from, so a thin average cannot look certain.

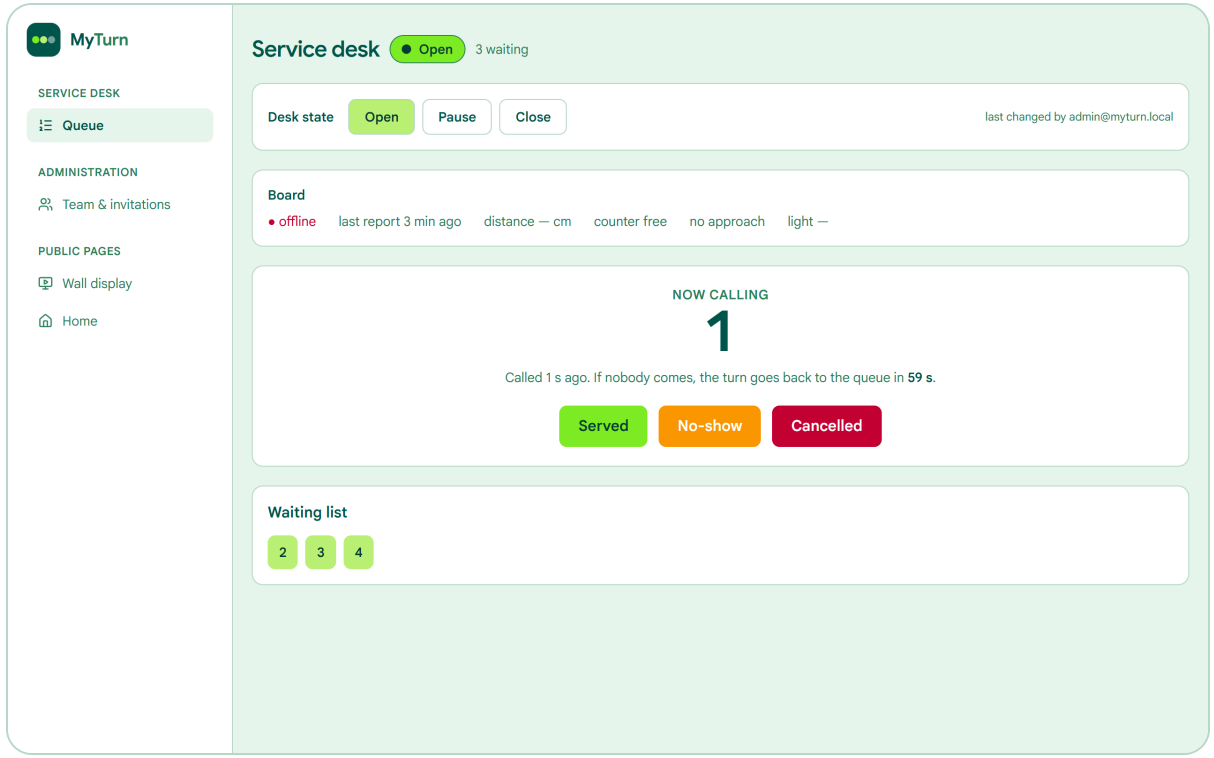
3 Small demo times are shown

A desk that serves in seconds reports seconds. The figure is not hidden to look better.

4 A target is compared, not promised

If the desk sets a target, the statistics page reports how often it was actually met.

Two views of the same queue.



Staff: call, complete, skip, cancel, walk-in, recall and undo.

Live: hold countdown, average service time, keyboard shortcuts.

Administration

A — Team & invitations
Invite, promote, switch off; the last administrator cannot be removed.

B — Activity
Ticket, desk and account history in one list, with the reason for each requeue.

C — Statistics
Counts, service-time distribution, busiest hours, target comparison and CSV.

D — Desk settings
Name, description and target time change while the server runs.

Every change is written to the activity log with the actor and the time.

Rights are granted, checked and recorded.

VISITOR

No account

An opaque ticket token in the link. No name, phone number or email is ever requested.

STAFF

Invitation

Single-use, 72 hours. Sign in as a person, or with the shared classroom password.

ADMINISTRATOR

Promoted

Created only by the seed script or by another administrator. Never by public registration.

DEVICE

Device key

The ESP32 sends a key with every reading. It can report and ask for a setting; it cannot change the queue.

Server-side roles. Every admin route is checked twice: the page, and the API.

Invitations cannot create admins. Asking for one is a 400.

No address leaks. A password reset answers the same whether or not the address exists.

CSRF on writes. A double-submit token, kept in memory, not in localStorage.

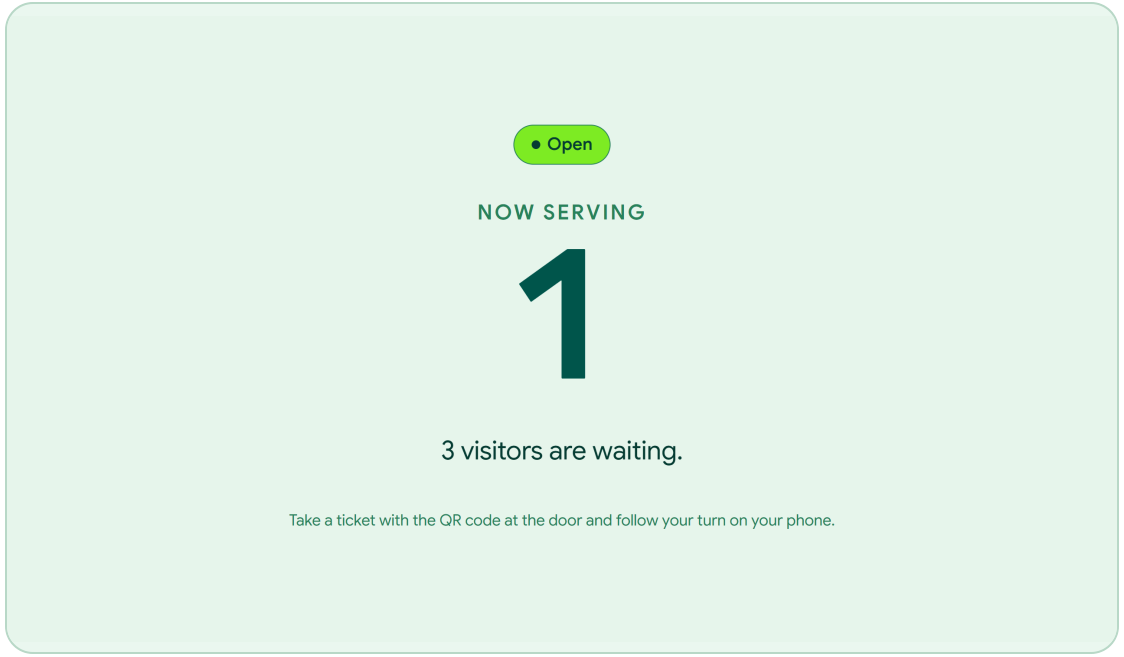
Guarded last admin. You cannot demote the final administrator or switch off your own account.

Recorded. Account changes land in the activity history with the actor and the time.

> A visitor never gets an account. A ticket asks for no name, phone number or email.

What we will show on the day.

1. **Take a ticket on a phone**
   The number comes from the server; the tracker opens.
2. **Watch the display change**
   Call next at the desk; the wall screen and the phone update together.
3. **Serve or record a no-show**
   Complete, skip or cancel. Undo is available for ten seconds.
4. **Let a call expire**
   The ticket goes to the back with a fresh number, and the reason is logged.
5. **Show the board**
   Distance and approach change the LED; the dashboard shows what the board applied.
6. **Close the desk**
   The ticket page says so before a visitor joins, and the LED follows.



The waiting-area display

Tested, not assumed.

172
API checks, including hold, requeue and estimate gate

71
browser checks on the visitor, auth and desk pages

45
console checks: activity, statistics, settings, roles

8
sidebar checks, including the shared-session rules

Every check was run against the running prototype, not read off the source.

✓ **The build is clean**
TypeScript, ESLint and the production build all pass; all fourteen pages are generated.

✓ **The rules are exercised**
Auth and CSRF, the occupancy requeue with synthetic readings, the estimate gate, undo, recall, CSV.

! **Still measured by hand**
The physical sensor trials and the photograph of the desk rig are the remaining figures; the report leaves them blank rather than inventing them.

> A feature is only finished when a check proves it.

What it does not do yet.

One desk, one service

The schema and the console assume a single desk. A second service needs a desk column and a per-desk call index.

CURRENT SCOPE

No separate SERVING state

A called ticket is treated as being served; the dwell time is the service time. This is documented, not invented.

DELIBERATE

Sensors measure only

The PIR can see a passer-by, and the ultrasonic sensor needs a clean mount. Field accuracy is still to be recorded.

TO MEASURE

Board offline means full hold

An offline board keeps the fixed 60-second hold. Readings are not buffered on the ESP32 during a Wi-Fi drop.

DESIGNED, NOT BUILT

No mail server

Invitation and reset links are shown in the app and logged in development, not emailed. A production deployment needs SMTP.

PROTOTYPE

Single process

SSE subscribers and rate-limit counters live in memory, so the API runs one worker. Scaling needs shared state.

DEPLOYMENT

Also not built: daily number reset, multi-desk routing, spoken announcements and a forecast. They are specified in the feature catalogue with their milestones.

Delivered, and what comes next.

DELIVERED

- ✓ A working prototype: 14 pages, 11 tables, real accounts and roles.
- ✓ The ESP32 sketch: distance, approach, LED and the desk setting.
- ✓ Four suites, 296 automated checks, all passing.
- ✓ A proposal, a 55-page report and the design documents.

NEXT

1. Run and record the physical sensor trials; add the desk photograph.
2. Fill the measured figures into the report, without estimating them.
3. Buffer readings on the board when the Wi-Fi drops.
4. Add a second desk and daily number resets.
5. Speak the number and show notices on the display.

A working prototype, a tested queue, and a clear list of what still needs measuring.

Thank you. Questions?

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